

Summary



TOPIC 4.1:

What is the universe, and how do we make sense of it?

- “Universe” is understood as different things to different people.
- The sense we make of the universe depends on who we are, when we are, and what we know.

Key Term

universe

ESSENTIAL QUESTION

How can we understand and explain the universe, its formation, and our place in it?



TOPIC 4.2:

What do we know about the universe based on what we can see only with our eyes?

- We see a variety of objects that appear in the sky in mostly predictable ways.
- The appearance and motions of the objects we see can be described using a model called the celestial sphere.

Key Terms

constellation

celestial sphere

ecliptic



TOPIC 4.3:

How has technology expanded our knowledge and understanding of the universe?

- We have invented telescopes and other devices that extend and enhance our sense of sight.
- We know that our Milky Way galaxy is just one of many billions of galaxies in the universe.
- There are vast distances separating stars and separating galaxies.
- The properties of stars help us develop an understanding of their life cycles.

Key Terms

galaxy	star cluster
light-year	Hertzsprung-Russell (H-R) diagram
main sequence	supernova
neutron star	black hole



TOPIC 4.4:

How do we use the big bang theory to explain what we know about the universe?

- The big bang theory is based on two main sets of evidence: redshift and cosmic background radiation.
- The big bang theory helps us describe how the components of the universe formed and have changed over time.
- There is much about the universe that we still cannot explain.

Key Terms

redshifted	blueshifted	big bang theory
cosmic microwave background (CMB) radiation		

Review

What Do You Know? Connecting to Concepts

Visualizing Key Ideas

1. Create a flowchart that outlines the life cycle for each of the following:
 - a) a low-mass star
 - b) an intermediate-mass star
 - c) a high-mass star
2. Design an infographic for a non-science audience that combines the information from question 1. Include any other information that you think is relevant.
3. Design a graphic organizer or infographic for the big bang. Your organizer or infographic should be the simplest possible representation that still communicates the most essential elements of the big bang. Someone looking at it should be able to understand its key ideas.
4. Flip through the pages of this unit. Locate and identify three images that you think are good answers to the following question: How does the idea that light takes time to travel across space give us an understanding that the universe is vast and ancient? Explain your choices. Then choose one as the best, and explain why you think it is.
5. The ancient Greeks called the constellation in the photo below Orion the Hunter. Sketch this constellation in your notebook.
 - a) Identify the name of any star that you know or recognize. Include the type of star it is, as well.
 - b) Identify the group of stars that make up Orion's belt and sword.



- c) Describe this constellation as it was seen by another nation (such as one belonging to First Peoples) or culture.
- d) Create your own constellation narrative for this grouping of stars.

Using Key Terms

5. Write a brief paragraph that connects the following terms: ecliptic, constellations, celestial sphere.
6. Create a table with three columns. In the first column, list all the terms relating to what we can view in the sky with our own eyes and also how we can locate them on the celestial sphere. In the second column, record a definition for each term, written in your own words. In the third column, draw a sketch that can combine as many of these terms as possible to show how they are interrelated.
7. Using a format of your choice, show how the term big bang is related to the following terms: blueshift, cosmic microwave background radiation, dark energy, dark matter, redshift.

Communicating Concepts

8. Explain why objects in the day and night sky appear to move as time passes each day.
9. What are two reasons that some stars are brighter than others when you look at the night sky?

10. The telescope near Penticton, B.C., is able to view stars during the day as well as at night. It also works on cloudy days and if it is snowing. Why is this possible?
11. Explain how the concept of the Doppler shift applies both to sound and to light, including the similarities and the differences.
12. When our star, the Sun, reaches the end of its life about 5 billion years from now, will it become a black hole? Why or why not?
13. The cosmic microwave background radiation is very cold, but it results from an event that was unimaginably hot. Explain how this is possible.
14. Explain how the positions of the stars and the Sun can be used for navigation.
15. a) The photo below shows a long-exposure image of the night sky. The bright tracings are called star trails. How does this photo provide evidence that stars in the northern hemisphere appear to revolve around a celestial object that remains stationary?
b) Name the stationary celestial object.



16. Why are the more massive stars the only important contributors in enriching the universe with elements heavier than hydrogen, helium, and lithium?
17. Why are white dwarf stars hot as well as dim?
18. Explain how the spectrum of light from a galaxy is useful to astronomers.

What Do You Know?

Connecting to Competencies

Developing Skills

19. Use at least two examples to describe how to use Polaris to navigate the night sky.
20. Explain how the rate of expansion of the distance between two points is proportional to that distance. Use a diagram or an example of an expandable surface, such as a balloon or rubber sheet, to illustrate your explanation. Then relate the rate of expansion to Hubble's law.
21. a) What is the difference between altitude and azimuth?
b) Describe two examples that illustrate how accuracy can be affected when you are using your hands to determine altitude and azimuth.
22. Nebulas can be formed by the explosion of supernovas. They are also the birth places of stars. Some have been given fanciful names based on how they appear in telescopes. Research a nebula that has been discovered using telescopes. Explain what it looks like, how it may have formed, how far away it is, what it is made of, and whether new stars are likely to form in it.
23. Explain why some stars on the Hertzsprung-Russell diagram are not located on the main sequence diagonal.
24. How did Indigenous people of the past depend on their knowledge of the skies for survival? Research to find out different ways that people applied their astronomical knowledge. Explain how these were tied to the places where they lived.



Unit 4 Review *(continued)*

Thinking Critically and Creatively

25. If the Earth did not rotate, could you still determine the ecliptic? Explain why, or why not.
26. Suppose that instead of seeing light, you perceived it as sound. In other words, suppose that you could “hear” light. What might it mean to hear different colours, and how would this change the way we might think of the cosmic redshift? (By the way, this kind of sensory ability—for example, to hear colour, smell sounds, or see flavours—has a name. Find out what it is. If this phenomenon interests you, investigate it further.)
27. Suppose you are an astronomer viewing a newly discovered galaxy. You record its spectrum and discover that some of its light is red-shifted, some of its light is blue shifted, and that most of its light is not shifted at all. What might be the cause of this effect? Could it be used to measure anything important about the galaxy?

Understanding Big Ideas Making New Connections

Applying Your Understanding

28. Suppose you are sailing from Vancouver to Hawaii and navigating only by compass and observations of the sky. You have star charts to refer to and instruments to make precise measurements. Explain why you would also need to have a precise clock or some method of telling time in order to navigate correctly.
29. Does the position of Polaris change if the latitude of the observer changes? Draw a sketch to explain your ideas. (Hint: Think about where you might look for Polaris if you were standing on the equator, or on the

North Pole. What is the relationship between the altitude of Polaris and the latitude of the observer.)

30. What does it mean when we say that much of what we understand about the universe has been learned by our ability to use light to look back in time.

Thinking Critically and Creatively

31. How does the structure of our solar system, in which planets orbit around the Sun, compare to the structure of the Milky Way galaxy? What in the galaxy corresponds to the solar system’s planets? to the Sun? to the motion of the objects in both and how both are held together?
32. Many First Peoples have words to describe a person who is trained to observe the skies. Sometimes they are translated as stargazer or moon reader. Find examples of such words in the language of a local First Nations community or other First Peoples cultures.



33. a) Describe what globular clusters are.
b) Explain how the properties of globular clusters were investigated and how the resulting discoveries led to further understanding of the Milky Way.
34. Propose an investigation you could use to investigate the relationship between the brightness or luminosity of a star and its distance from Earth. Outline the materials and equipment you would need, any safety considerations, the plan or procedure for the investigation, and the results you would expect to obtain. Provide a rationale for your plan and your expected results. (Limitation: You cannot use flashlights or laser pointers.)

- 35.** Explain the significance of the following observations.
- a)** Most stars that we see in the night sky from Earth are in a band that runs north-south across the sky.
 - b)** Globular clusters are all located in one direction in the sky and not all around Earth.
- 36.** The photo below shows a reconstruction of a device called the Antikythera (an-tih-keh-THEE-rah) mechanism. The incomplete remains of this technology, at least 2200 years old, were discovered in the year 1900 as part of a sunken shipwreck off the coast of the Greek island of Antikythera. It has been described as an ancient computer whose purpose was to model the motions of celestial objects.



- a)** Find at least three reputable, reliable sources of information about the Antikythera mechanism to learn more about it and how it has been reconstructed. Summarize your findings.
- b)** Explain how you evaluated a site to be able to consider it reputable and reliable. In other words, what criteria did you use to judge each site?

- 37.** The *Star Trek* “universe” invented warp drive to move through space at greater-than-light speed. Warp 9.9 is about 3000 times faster than light speed. Calculate how long it would take a starship with warp drive to reach
- a)** Proxima Centauri
 - b)** the centre of the Milky Way
 - c)** the region of the Andromeda Galaxy

Connecting to Self and Society

- 38.** At the time of writing this question, a plan to build the largest telescope in the world, one with a diameter of 30 metres, is being considered for the top of the highest mountain in Hawaii, Mauna Kea. Canada is part of this plan. This same site is considered a sacred place by Hawaiian First Peoples, and many there do not want the telescope built on this site. Research this issue and imagine you are to be a mediator to help the sides come to a positive resolution to the issue. What approach would you take to help bring resolution?
- 39.** The World Wide Web was invented, in part, to facilitate communication related to scientific investigations involving astronomers and physicists. How would your life be different if it had not been invented?
- 40.** Canadian astronaut Chris Hadfield was inspired to explore space by novels and comics he read in his youth. African-American astronaut Mae Jemison was inspired by the ethnic diversity of the characters in *Star Trek*. At the time, no similar diversity existed in the American space program or on TV. Today, astronauts from many different nations and ethnic backgrounds work upon the International Space Station. What role might science fiction have played in inspiring this diversity. Justify your response.



Unit Assessment

What are our responsibilities as explorers of the universe?



With our eyes and imaginations, and then assisted by technology such as telescopes, satellites, and probes, we humans have always been exploring the universe. Many nations around the globe have launched technologies, and ourselves, beyond Earth's atmosphere—in a few cases, such as the *Voyager* space probes, *far* beyond. What, if any, cosmic conscience are we obliged to have to balance or inform our powerfully human sense of curiosity?

Work as part of a group to do the following.

- STEP 1** ▶ Reflect on the three options, their photos, and the question asked for each option.
- STEP 2** ▶ Brainstorm at least three more options and questions of your own about a situation related to the unit assessment question.
- STEP 3** ▶ Decide on one of the six option questions to investigate.
- STEP 4** ▶ Plan and conduct a scientific inquiry to explore your question.
- STEP 5** ▶ Organize and analyze the data and information that you find and collect.
- STEP 6** ▶ Communicate the results of your inquiry in a suitable manner.

OPTION A

Protecting Life

How do we protect life on Earth from potential aliens as well as protect possible alien life from contamination by us?



OPTION B

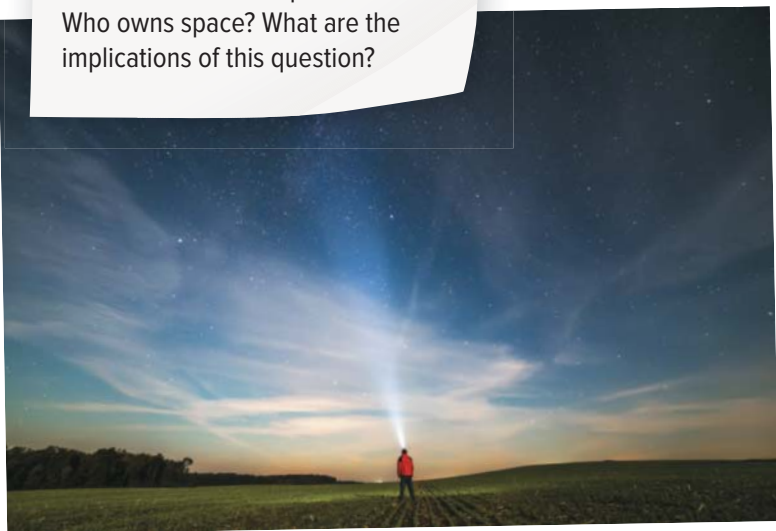
Societal Responsibility

Is it responsible to spend taxpayer money on space investigation when so many on Earth live in poverty?

OPTION C

Ethical Dilemmas

No single nation “owns” Antarctica or the International Space Station. Who owns space? What are the implications of this question?



Assessment Criteria

Did I and my group...

- Develop one or more questions that provided opportunities for rich investigation? **QP**
- Develop effective methods to collect and record reliable data and information? **PC**
- Apply different ways of knowing to analyze, reflect on, and draw meaningful conclusions that are consistent with evidence? **PA**
- Consider and demonstrate an awareness of assumptions, bias, and social, ethical, and environmental implications over the whole process of our inquiry? **E**
- Propose alternative courses of thought and/or action that contribute to care for self, others, community, and world? **AI**
- Construct evidence-based arguments using language, conventions, and representations appropriate for a specific purpose and audience? **C**